A Memoir on Cubic Surfaces.

By Professor Cayley.

[Continued from Vol. VI, pp. 259–366.]

Section I = 12. Article Nos. 38 to 46. (Continued.)

Double-sixers.

It has been mentioned that the number of double-sixers was = 36; these are as follows:

$1, 2, 3, 4, 5, 6$	Assumed primitive	
$1', 2', 3', 4', 5', 6'$		
$1, 1', 23, 24, 25, 26$	Like arrangements	15
$2, 2', 13, 14, 15, 16$		
$1, 2, 3, 56, 46, 45$	Like arrangements	20
$23, 13, 12, 4, 5, 6$		
	Total	36

where, if we take any column of two lines, we have the complete number 216 of pairs of non-intersecting lines (each line meets 10 lines, there are therefore $27 - 1 - 10 = 16$ lines which it does not meet; $27 \times 16 \div 2 = 216$).

Dr. Hart's Arrangement.

Dr. Hart arranges the 27 lines, cubically, thus:

$$\begin{array}{ccccccccc} A_1 & B_1 & C_1 & a_1 & b_1 & c_1 & \alpha_1 & \beta_1 & \gamma_1 \\ A_2 & B_2 & C_2 & a_2 & b_2 & c_2 & \alpha_2 & \beta_2 & \gamma_2 \\ A_3 & B_3 & C_3 & a_3 & b_3 & c_3 & \alpha_3 & \beta_3 & \gamma_3 \end{array}$$

where letters of the same alphabet denote lines in the same plane, if only the letters are the same or the suffixes the same; thus A_1, A_2, A_3 lie in a plane $A_1A_2A_3$; A_1, B_1, C_1 lie in a plane $A_1B_1C_1$. Letters of different alphabets denote lines which meet according to the Table.

Equations of Planes.

Using the notation where $[12'] = w$, $[23'] = \theta$, etc., the equations of the planes are:

$$
\begin{aligned}
W &= 0, && [12' = w] \\
X &= 0, && [42' = y] \\
Y &= 0, && [14' = z] \\
Z &= 0, && [12' = w] \\
\frac{X}{l} + \frac{Y}{m} + \frac{Z}{n} + W &= 0, && [41' = \xi] \\
\frac{X}{l} + mY + \frac{Z}{n} + W &= 0, && [34' = \eta] \\
\frac{X}{l} + \frac{Y}{m} + nZ + W &= 0, && [13.24.56 = h] \\
lX + mY + nZ + W &= 0, && [24' = \zeta] \\
lX + \frac{Y}{m} + nZ + W &= 0, && [14.25.36 = g] \\
lX + mY + \frac{Z}{n} + W &= 0, && [43' = \xi] \\
\frac{l(p-\alpha) + 2mn}{p+\alpha} X + \cdots &= 0, && [52' = \psi] \\
\frac{n(p-\alpha) + 2lm}{p+\alpha} Y + \cdots &= 0, && [15' = z] \\
\frac{m(p-\alpha) + 2nl}{p+\alpha} Z + \cdots &= 0, && [12.36.45 = \chi]
\end{aligned}
$$

Further Plane Equations.

Continuing the equations of the planes:

$$
\begin{aligned}
-m\left\{\frac{n(p-\alpha)}{\cdots}\right\} X + mY + nZ + W &= 0, && [56' = l] \\
lX - m\left\{\frac{l(p-\alpha)}{\cdots}\right\} Y + nZ + W &= 0, && [45' = m] \\
lX + mY - n\left\{\frac{m(p-\alpha)}{\cdots}\right\} Z + W &= 0, && [64' = n] \\
\frac{l(p-\alpha)}{\cdots} X + mY + \frac{Z}{n} + W &= 0, && [15.26.34 = l_1] \\
lX + \frac{2n}{\cdots} Y + nZ + W &= 0, && [16.24.35 = m_1] \\
lX + mY - \frac{m(p-\alpha)}{\cdots} Z + W &= 0, && [14.25.36 = n_1] \\
\frac{2n}{\cdots} X + mY + \frac{Z}{n} + W &= 0, && [65' = l_1] \\
\frac{X}{l} + \frac{2n}{\cdots} Y + nZ + W &= 0, && [46' = m_1] \\
\frac{X}{l} + mY - \frac{m(p-\alpha)}{\cdots} Z + W &= 0, && [54' = n_1] \\
\frac{X}{l} + mY + \frac{Z}{n} + W &= 0, && [16.25.34 = l_1] \\
\frac{2l}{\cdots} X + mY + nZ + W &= 0, && [15.24.36 = m_1] \\
lX - \frac{2l}{\cdots} Y + nZ + W &= 0, && [14.26.35 = n_1]
\end{aligned}
$$

Complex Plane Equations.

The remaining plane equations involve more complex coefficients:

$$-\frac{2l}{\cdots}X + nY + mZ + \{mn(p-\alpha) - 2l(l^2-m^2-n^2)\}W = 0,$$

$$[36' = p_1]$$

$$mX - \frac{2m}{\cdots}Y + lZ - \frac{l}{\cdots}\{m(1-n^2)(p-\alpha) - 2nl\}W = 0,$$

$$[25' = q_1]$$

$$nX + lY - \frac{2n}{\cdots}Z - \frac{l}{\cdots}\{n(1-l^2-m^2)(p-\alpha) - 2lm\}W = 0,$$

$$[13.26.45 = r_1]$$

Coordinates of the 27 Lines.

The coordinates of the 27 lines are then found to be as follows:

(a) (b) (c)

1 :	1,	1,	1,	$1 - k\left\{\frac{l}{m}\right\}\left\{\frac{m}{n}\right\}\cdots$
2 :	1,	$-k\left\{\frac{l}{m}\right\}\left\{\frac{m}{n}\right\}$,	1,	$1\cdots$
3 :	$-k(l-1)(m-1)$,	$-k(l-1)(m-1)$,	1,	$\cdots$

(f) (g) (h)

$(12 = a_1)$:	$-n$,	m,	1,	$-l\cdots$
$(1' = b_2)$:	1,	$-l$,	n,	$m\cdots$
$(23 = b_1)$:	$-m$,	l,	1,	$n\cdots$
$(3' = c_1)$:	1,	1,	$-l$,	$m\cdots$

The remaining line coordinates follow similar patterns with appropriate permutations of the parameters l, m, n and their combinations.

Equation of the Cubic Surface.

We have $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ for the equations of the planes $(12.34.56 = x)$, $(42' = y)$, $(14' = z)$, $(12' = w)$; and representing by $\xi = lX + \frac{Y}{m} + \frac{Z}{n} + W = 0$ the equation of any other plane

$(41' = \xi)$, the equation of the cubic surface may be presented in the several forms:

$$
\begin{aligned}
&= Fgg_1 + k\eta\zeta ZX,\\
&= Fhh_1 + k\zeta\theta XY,\\
&= W\eta\eta_1 + k\theta\iota YZ,\\
&= W\zeta\zeta_1 + k\iota\kappa ZX,\\
&= W\theta\theta_1 + k\kappa\lambda XY,\\
&= W\iota\iota_1 + k\lambda\mu YZ,\\
&= W\kappa\kappa_1 + k\mu\nu ZX,\\
&= W\lambda\lambda_1 + k\nu\xi XY,\\
&= W\mu\mu_1 + k\xi\eta YZ,\\
&= W\nu\nu_1 + k\xi\eta\zeta,\\
&= W\xi\xi_1 + k\eta\zeta\iota,\\
&= W\pi\pi_1 + k\xi\eta z,\\
&= W\rho\rho_1 + k\zeta\theta z,\\
&= W\sigma\sigma_1 + k\xi\iota x.
\end{aligned}
$$

Section II = 12–C_5. Article Nos. 47 to 59.

The Reciprocal Surface.

It may be remarked that the system of lines and planes is at once deduced from that belonging to I = 12, by supposing that in the double-sixer the lines $1, 2, 3, 4, 5, 6$ become the rays of a cubic node C_5. The reciprocal of a double-sixer is a double-sixer. Hence the 27 lines of the reciprocal surface may be (and that in 36 different ways) represented by:

$$
\begin{array}{cccccc}
1, & 2, & 3, & 4, & 5, & 6\\
1', & 2', & 3', & 4', & 5', & 6'\\
12, & 13, & \dots & 56, & &
\end{array}
$$

where 12 is now the line joining the points $12'$ and $1'2$; and so for the other lines. The lines $12, 34, 56$ meet in a point 12.34.56; the 30 points $12', 1'2, \dots, 56', 5'6$, and the fifteen points 12.34.56 make up the 45 points t'.

The above equation, $S^3 - T^2 = 0$, shows that the cuspidal curve is a complete intersection 6×4; $c' = 24$.

Equation $W(a, b, c, f, g, h \setminus X, Y, Z)^2 + 2kXYZ = 0$.

The system of lines and planes is at once deduced from that belonging to II $= 12 - C_5$, by supposing the tangent cone to reduce itself to the pair of biplanes.

The Diagram.

Lines.	
12	
13	
14	
15	
16	
23	Biradial planes, through each pair of rays. $15 \times 2 = 30$
24	
25	
26	
34	
35	
36	
45	
46	
56	
12.34.56	
12.35.46	
12.36.45	Planes each containing three non-intersecting lines. $15 \times 1 = 15$
13.24.56	
13.25.46	
13.26.45	
14.23.56	
14.25.36	
14.26.35	
15.23.46	
15.24.36	
15.26.34	
16.23.45	
16.24.35	
16.25.34	
30	45

Equations of Planes.

Putting the equation of the surface in the form:

$$W\,\{1,1,1,1 \,\backslash\, X,Y,Z\}^2 + \frac{2}{P}XYZ = 0,$$

where for shortness:

$$\lambda = mn - l, \quad \mu = nl - m, \quad \nu = lm - n, \quad \delta = lmn - 1, \quad \rho = lmn,$$

then taking $X=0$ as the equation of the plane [12], $Y=0$ as that of the plane [34], $Z=0$ as that of the plane [56], the equations of the 30 distinct planes are found to be:

$$\begin{aligned}
X&=0, &&[12]\\
Y&=0, &&[34]\\
Z&=0, &&[56]\\
mX+lY+Z&=0, &&[23]\\
m^{-1}X+Y+Z&=0, &&[24]\\
X+nY+mZ&=0, &&[45]\\
mX+l^{-1}Y+Z&=0, &&[13]\\
m^{-1}X+l^{-1}Y+Z&=0, &&[14]\\
X+n^{-1}Y+mZ&=0, &&[46]
\end{aligned}$$

Coordinates of the 21 Distinct Lines.

And the coordinates of the 21 distinct lines are:

(a) (b) (c) (f) (g) (h) whence equations may be taken to be:

1 :	l,	-1,	1,	(1) $X=0$, $Y+lZ=0$
2 :	1,	-1,	1,	(2) $Y=0$, $Z+mX=0$
3 :	n,	-1,	1,	(3) $Z=0$, $X+nY=0$
4 :	1,	-1,	1,	(4) $X=0$, $Y+nZ=0$
5 :	1,	-1,	1,	(5) $Y=0$, $Z+lX=0$
6 :	m,	n,	1,	(6) $Z=0$, $X+m^{-1}Y=0$

And for the remaining lines:

(45) :	$\frac{1}{n}$,	$\frac{1}{m}$,	1,	$\frac{\alpha}{\gamma}$,	$\frac{1}{\gamma}$,	$\frac{1}{\beta}$
(16) :	$\frac{1}{n}$,	$\frac{1}{l}$,	1,	$\frac{\gamma}{\beta}$,	$\frac{1}{\beta}$,	$\frac{1}{\alpha}$
(23) :	$\frac{1}{m}$,	l,	1,	$\frac{\beta}{\alpha}$,	$\frac{1}{\alpha}$,	$\frac{1}{\gamma}$
(46) :	n,	m,	1,	$\frac{1}{\gamma}$,	$\frac{\alpha}{\gamma}$,	$\frac{1}{\beta}$
(26) :	l,	m,	1,	$\frac{1}{\beta}$,	$\frac{\gamma}{\beta}$,	$\frac{1}{\alpha}$

The Hessian Surface.

Resuming the equation $W(a,b,c,f,g,h \setminus X,Y,Z)^2+2kXYZ=0$, the equation of the Hessian surface is found to be:

$$\begin{aligned}
&kW^2(a,b,c,f,g,h \setminus X,Y,Z)^2\\
&\qquad +2kW\{(a,b,c,f,g,h \setminus X,Y,Z)^2(FX+GY+HZ)-k^2XYZ\}\\
&\qquad -k\{a^2X^4+b^2Y^4+c^2Z^4-2bcY^2Z^2-2caZ^2X^2-2abX^2Y^2\\
&\qquad\qquad -4XYZ[(af+gh)X+(bg+hf)Y+(ch+fg)Z]\}=0,
\end{aligned}$$

where

$$(A,B,C,F,G,H)=(bc-f^2,\ ca-g^2,\ ab-h^2,\ gh-af,\ hf-bg,\ fg-ch),$$

and

$$K=abc-af^2-bg^2-ch^2+2fgh.$$

Invariants and Reciprocal Surface.

Write as before (A, B, C, F, G, H) for the inverse coefficients ($A = bc - f^2$, &c.), and $K = abc - af^2 - bg^2 - ch^2 + 2fgh$; and moreover:

$$\begin{aligned}
\Phi &= (A, B, C, F, G, H \setminus x, y, z)^2,\\
P &= Ax + Hy + Gz,\\
Q &= Hx + By + Fz,\\
R &= Gx + Fy + Cz,\\
t &= fx + gy + hz,\\
U &= afyz + bgzx + chxy,\\
V &= 2Kxyz - aPyz - bQzx - cRxy\\
&= -aHy^2z - bFz^2x - cGx^2y - aGyz^2 - bEzx^2 - cFxy^2\\
&\quad + (-abc - ap^2 - bg^2 - ch^2 + 4fgh)xyz,\\
W &= (A, B, C, F, G, H \setminus ayz, bzx, cxy)^2,\\
L &= kW - 2ktw - \Phi,\\
M &= kwU + V,\\
N &= 2kabcxyzw + W.
\end{aligned}$$

Then the invariants of the ternary cubic are:

$$\begin{aligned}
S &= L^2 - 12kwM,\\
T &= L^3 - 18kwLM - 54k^2w^2N,
\end{aligned}$$

and the required equation of the reciprocal surface is:

$$\frac{1}{k^2w^2}\left[(L^3 - 18kwLM - 54k^2w^2N)^2 - (L^2 - 12kwM)^3\right] = 0,$$

viz. this is:

$$0 = DN = (kW - 2ktw - \Phi)^2(2kabcxyzw + W) + L^2M^2 + \cdots$$

Section III $= 12$–B_3. Article Nos. 60 to 72.

Equation $2W(X + Y + Z)(lX + mY + nZ) + 2kXYZ = 0$.

The system of lines and planes is at once deduced from that belonging to II $= 12 - C_5$, by supposing the tangent cone to reduce itself to the pair of biplanes: three of the planes (α) of II $= 12 - C_5$ thus coming to coincide with the one biplane, and three of them with the other biplane.

The Diagram.

Lines.	
$\mathrm{III} = 12 - B_3$	
1	
2	
$1'$	
$2'$	
12	Biplanes containing rays $1, 2$ and $1', 2'$ respectively. $2 \times 12 = 24$
$11'$	
$22'$	Plane touching along edge and containing the transversal. $1 \times 8 = 8$
$12'$	
$21'$	Biradial planes each containing a ray of the one and a ray of the other biplane. $4 \times 4 = 16$
$11'.23'$	
$12'.31'$	Planes each through the transversal. $2 \times 1 = 2$
45	
46	
56	
3	
4	
5	
6	
$3'$	
$4'$	
$5'$	
$6'$	

Equations of Planes.

Taking $X + Y + Z = 0$ for the biplane that contains the rays $1, 2, 3$, and $lX + mY + nZ = 0$ for that which contains the rays $4, 5, 6$, we may take $X = 0$, $Y = 0$, $Z = 0$ for the equations of the planes [14], [25], [36] respectively; and then writing for shortness:

$$m - n = \lambda, \quad n - l = \mu, \quad l - m = \nu,$$

and assuming, as we may do, $k = \sqrt{\lambda\mu\nu}$, so that the equation of the surface is:

$$W(X + Y + Z)(lX + mY + nZ) + (m - n)(n - l)(l - m)XYZ = 0,$$

the equations of the 17 distinct planes are:

$$\begin{aligned} X &= 0, & [14] \\ Y &= 0, & [25] \\ Z &= 0, & [36] \\ X + Y + Z &= 0, & [123] \\ lX + mY + nZ &= 0, & [456] \end{aligned}$$

Coordinates of the 15 Distinct Lines.

And the coordinates of the fifteen distinct lines are:

(a) (b) (c) (f) (g) (h) whence equations may be written:

(1) :	$-1,$	$1,$	$1,$	(1) $X = 0, \quad Y + Z = 0$
(2) :	$1,$	$-1,$	$1,$	(2) $Y = 0, \quad Z + X = 0$
(3) :	$1,$	$1,$	$-1,$	(3) $Z = 0, \quad X + Y = 0$
(1′) :	$-n,$	$m,$	$1,$	(4) $X = 0, \quad mY + nZ = 0$
(2′) :	$n,$	$-l,$	$1,$	(5) $Y = 0, \quad nZ + lX = 0$
(3′) :	$l,$	$m,$	$-1,$	(6) $Z = 0, \quad lX + mY = 0$
(14) :	$1,$	$-1,$	$1,$	(14) $X = 0, \quad W = 0$
(25) :	$1,$	$1,$	$-1,$	(25) $Y = 0, \quad W = 0$
(36) :	$-1,$	$1,$	$1,$	(36) $Z = 0, \quad W = 0$

And the remaining lines:

(15) :	$l,$	$n,$	$n,$	$n^2\nu - n\lambda\mu$
(16) :	$l,$	$m,$	$m,$	$m^2\mu - lm\nu$
(26) :	$l,$	$m,$	$l,$	$l^2\lambda - lm\mu$
(24) :	$n,$	$m,$	$n,$	$n^2\nu - mn\lambda$
(34) :	$m,$	$m,$	$n,$	$m^2\mu - mn\nu$
(35) :	$l,$	$l,$	$n,$	$l^2\lambda - ln\nu$

The Rays and Facultative Lines.

The rays are not, the mere lines are, facultative; hence $b' = p' = 6$; $t' = 6$.

The Hessian Surface.

The equation of the Hessian surface is:

$$- W(X+Y+Z)(lX+mY+nZ)(\nu X+\lambda Y+\mu Z) \\ - k(l^2X^2+m^2Y^2+n^2Z^2-2mnYZ-2nlZX-2lmXY) = 0.$$

Section IV $= 12$–B_4. Article Nos. 73 to 84.

Equation $ZW(e, X-Y)(e_2X-Y)(e_3X-Y)(e_4X-Y)+2kXYZ=0$.

Writing $Z(a,b,c,d \setminus X,Y)^2 \cdot d_1X-Y \cdot d_2X-Y \cdot d_3X-Y \cdot d_4X-Y$, the 20 planes are:

$$
\begin{aligned}
Z &= 0, & [0]\\
X-f_1Y &= 0, & [11']\\
X-f_2Y &= 0, & [22']\\
X-f_3Y &= 0, & [33']\\
X-f_4Y &= 0, & [44']\\
Z-(f_1+f_2)Y-f_1f_2Z &= 0, & [12]\\
Z-(f_1+f_3)Y-f_1f_3Z &= 0, & [13]\\
Z-(f_1+f_4)Y-f_1f_4Z &= 0, & [14]\\
Z-(f_2+f_3)Y-f_2f_3Z &= 0, & [23]\\
Z-(f_2+f_4)Y-f_2f_4Z &= 0, & [24]\\
Z-(f_3+f_4)Y-f_3f_4Z &= 0, & [34]\\
Z-(f_1+f_2)Y-f_1f_2Y &= 0, & [1'2']\\
Z-(f_1+f_3)Y-f_1f_3Y &= 0, & [1'3']\\
Z-(f_1+f_4)Y-f_1f_4Y &= 0, & [1'4']\\
Z-(f_2+f_3)Y-f_2f_3Y &= 0, & [2'3']\\
Z-(f_2+f_4)Y-f_2f_4Y &= 0, & [2'4']\\
Z-(f_3+f_4)Y-f_3f_4Y &= 0, & [3'4']
\end{aligned}
$$

and the planes through the nodes:

$$
\begin{aligned}
(f_1+f_2+f_3+f_4)Z+\cdots &= 0, & [12.34]\\
(f_1+f_3+f_2+f_4)Z+\cdots &= 0, & [13.24]\\
(f_1+f_4+f_2+f_3)Z+\cdots &= 0, & [14.23]
\end{aligned}
$$

The 16 Lines.

And the 16 lines are:

(a) (b) (c) (f) (g) (h) whence equations may be written:

(0) :	1,	0,	0,	$x=0, \quad Y=0$
(5) :	0,	$-\gamma$,	d,	$X=0, \quad dY+\gamma Z+dW=0$
(1) :	1,	e_1,	0,	$e_1X-Y=0, \quad Z=0$

and so for the other lines with appropriate substitutions of e_2, e_3, e_4.

Section V $= 12$–B_3. Article Nos. 85 to 94.

Equation $WXZ+(X+Z)(Y^2-aX^2-bZ^2)=0$.

The diagram of the lines and planes is:

Lines.

V $= 12 - B_3$	
1	
2	
$1'$	
$2'$	Biplanes containing rays $1, 2$ and $1', 2'$ respectively. $2 \times 12 = 24$
12	
$11'$	Plane touching along edge and containing the transversal. $1 \times 8 = 8$
$22'$	
$12'$	Biradial planes each containing a ray of the one and a ray of the other biplane. $4 \times 4 = 16$
$21'$	
$11'.23'$	Planes each through the transversal. $2 \times 1 = 2$
$12'.31'$	
3	
4	
5	
6	
$3'$	
$4'$	
$5'$	
$6'$	

The Planes.

The planes are:

$$
\begin{aligned}
Z = 0, &\quad [12] \\
X = 0, &\quad [1'2'] \\
Y + \sqrt{a}X + \sqrt{b}Z = 0, &\quad [11'] \\
Y - \sqrt{a}X - \sqrt{b}Z = 0, &\quad [22'] \\
Y + \sqrt{a}X - \sqrt{b}Z = 0, &\quad [12'] \\
Y - \sqrt{a}X + \sqrt{b}Z = 0, &\quad [21']
\end{aligned}
$$

with other planes determined by the appropriate combinations.

Section VI = 12–B_4. Article Nos. 95 to 107.

Equation $Z(X-Y)(\theta_1 X - Y)(\theta_2 X - Y)(\theta_3 X - Y)(\theta_4 X - Y) = 0$.

Writing $(a, b, c, d \setminus Z, Y)^2 \cdot d_1 X - Y \cdot e_1 X - Y \cdot e_2 X - Y \cdot e_3 X - Y \cdot e_4 X - Y$, the planes are:

$$
\begin{aligned}
X &= 0, & [0]\\
Z &= 0, & [00]\\
\theta_1 X - Y &= 0, & [22']\\
\theta_2 X - Y &= 0, & [33']\\
\theta_3 X - Y &= 0, & [44']\\
\theta_4 X - Y &= 0, & [55']\\
d(e_1 X - Y) - Z &= 0, & [12]\\
d(e_2 X - Y) - Z &= 0, & [13]\\
d(e_3 X - Y) - Z &= 0, & [14]\\
X\theta_1 - Y(\theta_2 + \theta_3) - W &= 0, & [2'3']\\
X\theta_2 - Y(\theta_1 + \theta_4) - W &= 0, & [2'4']\\
X\theta_3 - Y(\theta_1 + \theta_4) - W &= 0, & [3'4']
\end{aligned}
$$

The Lines.

And the lines are:

(a) (b) (c) (f) (g) (h) equations may be written:

(0) :	1,	0,	0,	$X = 0, \quad Y = 0$
(1) :	-1,	d,	1,	$X = 0, \quad dY + Z = 0$
(2) :	1,	e_1,	1,	$e_1 X - Y = 0, \quad Z = 0$
(3) :	1,	e_2,	1,	$e_2 X - Y = 0, \quad Z = 0$
(4) :	1,	e_3,	1,	$e_3 X - Y = 0, \quad Z = 0$

and for the remaining lines, the coordinate expressions are more conveniently written in the symmetrical form.

The mere lines are facultative; hence $b' = p' = t' = \ldots$

Section VII = 12–$2C_2$. Article Nos. 108 to 116.

Equation $Y^2 + Y(X + Z + W) + 4\phi XZW = 0$.

Each of the lines $(X - Y = 0, Y + Z = 0)$ and $(X + W = 0, Y - Z = 0)$ is a double tangent of the spinode octic; in fact for the first of these lines we have:

$$16\theta^4 + 8\theta^2 + 16\phi\theta^2 + 5 = 0, \quad 16\theta^4 + 16\theta^2 + 4\phi = 0,$$

that is:

$$(2\theta^2 + 1)(8\theta^2 - 4\theta + 5) = 0, \quad 4\theta(2\theta^2 + 1) = 0,$$

so that the line touches at the two points given by $2\theta^2 + 1 = 0$; and similarly the other line touches at the two points given by $2\theta^2 - 1 = 0$.

The edge $X = 0$, $Z = 0$ has apparently a higher contact with the spinode octic, viz. the equations $X = 0$, $Z = 0$ are satisfied by $\theta = \infty$ twice, $\theta = 0$ five times; but it must be reckoned only as a double tangent. Hence $\beta' = 2 \cdot 2 + 2 = 6$.

Reciprocal Surface.

The equation is obtained by equating to zero the discriminant of the binary quartic:

$$Z^2(yZ - wX)^2 + 4wZ^2(wZ^2 + zZX + xX^2),$$

viz. calling $t = \dots$

Section VIII = 12–$3C_2$. Article Nos. 117 to 125.

Equation $Y^2 + Y(X + Z + W) + 4\phi XZW = 0$.

The diagram of the lines and planes is:

Lines.

VIII $= 12 - 3C_2$	
1	
7	Planes each touching along an axis and containing the corresponding transversal. $3 \times 2 = 6$
8	
9	
12	
34	Biradial planes each containing two rays of the same node. $3 \times 2 = 6$
56	
13	
24	Planes each containing an axis, and two rays through the terminal nodes respectively. $6 \times 4 = 24$
16	
26	
46	
35	
789	Plane through the three axes. $1 \times 8 = 8$
789	Plane through the three transversals. $1 \times 1 = 1$
14	
45	

[End of Memoir on Cubic Surfaces, pp. 401–450 of Vol. VI.]